

### Actinobacteria

|                     |                   |                    |
|---------------------|-------------------|--------------------|
| Taxa-Mixed model    | 0.9<br>$\pm 0.02$ | 0.84<br>$\pm 0.09$ |
| Taxa-Specific model | 0.9<br>$\pm 0.02$ | 0.81<br>$\pm 0.06$ |
|                     | Taxon-Specific    | Other-Taxa         |

### Firmicutes

|                     |                    |                    |
|---------------------|--------------------|--------------------|
| Taxa-Mixed model    | 0.75<br>$\pm 0.17$ | 0.89<br>$\pm 0.04$ |
| Taxa-Specific model | 0.75<br>$\pm 0.17$ | 0.8<br>$\pm 0.08$  |
|                     | Taxon-Specific     | Other-Taxa         |

### Bacteroidetes

|                     |                        |                    |
|---------------------|------------------------|--------------------|
| Taxa-Mixed model    | 0.88<br>$\pm 0.09$     | 0.85<br>$\pm 0.06$ |
| Taxa-Specific model | 0.88<br>$\pm 0.09$     | 0.83<br>$\pm 0.04$ |
|                     | Taxon-Specific dataset | Other-Taxa dataset |

### Proteobacteria

|                     |                        |                    |
|---------------------|------------------------|--------------------|
| Taxa-Mixed model    | 0.87<br>$\pm 0.05$     | 0.84<br>$\pm 0.09$ |
| Taxa-Specific model | 0.86<br>$\pm 0.06$     | 0.77<br>$\pm 0.06$ |
|                     | Taxon-Specific dataset | Other-Taxa dataset |